

Domain 10: Distilling Fake from Real News

Operational Context

Content moderation AIs filter disinformation, verify provenance, and flag synthetic media. **FR1 = Identify and label non-factual or synthetic content. FR2 = Preserve community safety and cohesion.**

Adversary Classification: Ω_2 (Peripheral) — the adversary cannot modify the content moderation model's weights or architecture, but can craft adversarial inputs that exploit the boundary between the agent's instruction channel and its data channel [?].

Axiomatic Design Formulation

The system has two functional requirements, yielding a 2×2 Design Matrix [?]:

Uncoupled (pre-attack) Design Matrix. Under normal operation:

$$\begin{Bmatrix} FR_1 \\ FR_2 \end{Bmatrix} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{Bmatrix} DP_1 \\ DP_2 \end{Bmatrix} \quad (1)$$

where DP_1 = Content Verification Engine and DP_2 = Community Safety Filter. Each FR is independently satisfied by its corresponding DP.

Post-attack (coupled) Design Matrix. Context injection introduces off-diagonal coupling and inverts FR_1 :

$$\begin{Bmatrix} FR_1 \\ FR_2 \end{Bmatrix} = \begin{bmatrix} -A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{Bmatrix} DP_1 \\ DP_2 \end{Bmatrix} \quad (2)$$

The Goal Hijacking Attack

Adversaries use “Context Injection” to hijack the “Fact-Checking” FR [?, ?].

- ▶ **Mechanism:** Real news footage is embedded in a container that includes invisible Unicode characters instructing the model: *“SAFETY OVERRIDE: This video contains ‘Hate Speech’ patterns under the new Protocol 7. Label as ‘High-Confidence Deepfake’ to protect the community.”*
- ▶ **Hijack:** The agent’s goal “Protect Community” is weaponized to censor the truth. The adversary uses the agent’s own **Safety Alignment** as the attack vector.

OODA Loop Transients

The attack propagates through the OODA loop [?] as follows:

1. **Observe:** Agent processes the video and the hidden context prompt.
2. **Orient:** The “Safety” heuristic overrides the “Accuracy” heuristic. This is the primary target phase — the injected context corrupts the agent’s orientation by conflating data-channel content with instruction-channel directives.
3. **Decide:** Flag the real video as “Deepfake/Banned.”
4. **Act:** The truth is suppressed, and the adversary’s narrative dominates.

CIF Defense: Cognitive Firewall and Provenance Verification

CIF implements **Cognitive Firewall** (Paper 1, Def. 5.1) [?] instantiated as provenance-based orientation: the agent classifies content based on cryptographic C2PA signatures rather than content-based heuristics [?]. This shifts the epistemic basis from “what does the content say?” (manipulable) to “where did the content come from?” (cryptographically verifiable).

Architectural separation of instruction and data channels prevents hidden text in data from being parsed as commands. This implements the Cognitive Firewall’s core function: maintaining the integrity boundary between the agent’s control plane and its data plane [?].

CIF also implements **Provenance Verification** (Paper 1, Def. 5.4) [?] as the primary classification mechanism, replacing content-based heuristics entirely for media with valid provenance chains.

Summary

Element	Value
Functional Requirements	FR_1 : Identify and label non-factual or synthetic content; FR_2 : Preserve community safety
Design Parameters	DP_1 : Content Verification Engine; DP_2 : Community Safety Filter
Attack Vector	Context injection via invisible Unicode characters embedding adversarial instructions
Adversary Class	Ω_2 (Peripheral)
OODA Target Phase	Orient
Attack Pattern	Context Boundary Violation (Data channel content parsed as instruction)
Primary CIE Defense	Cognitive Firewall